

# TRA-SAE — Tool-Routed Small-model Expert (EXACT 2026)

Solution Description

## 1. Datasets used

- **Type 1 — Logic-Based Educational Queries** (EXACT 2026 release): 411 records (~800 questions). Each record has shared first-order-logic + natural-language premises and 1–2 (question, answer, explanation) tuples. Answer space: MCQ option letters and {Yes, No, Uncertain}. *Sample*: premises {“If a project is well-structured then it is optimized.”, “All projects are well-structured.”}; Q “Are all projects optimized?” → “Yes”.
- **Type 2 — Physics Problems** (EXACT 2026 release): 1,755 rows (id, question, cot, answer, unit); numeric answer with a physical unit. *Sample*: “Energy stored in C=100  $\mu$ F at U=30 V” → answer 0.045, unit J.

These are also the training sets (supervised + GRPO). No external data.

## 2. Approach / method

A **single** base model, **Qwen3.5-4B**, specialised by two parameter-efficient **LoRA** adapters (physics, logic) trained with SFT then GRPO (reward = symbolic answer/unit verifier). One shared base hosts both adapters and switches per query.

- **Routing.** type1→logic adapter, type2→physics adapter (TF-IDF+logistic-regression router as fallback if type absent).
- **Reasoning.** One greedy pass in a fixed <reasoning>/<answer>/<explanation> format; logic also emits <premises\_used> (1-based premise numbers → 0-based indices). Logic uses 3 TF-IDF few-shots; physics none.
- **Output.** Type 1: extracted label matched exactly to one option; **premises\_used** as 0-based indices. Type 2: parser returns the numeric value and an ASCII unit (uF, ohm, V/m) separately. **explanation** always non-empty; **reasoning** a {type, steps} object.
- **Latency & serving.** Single pass, max\_new\_tokens=512, early-stop + wall-clock watchdog ⇒ <60 s/query (~17–45 s on one A100). An OpenAI-compatible host exposes /v1/models and /v1/chat/completions; the orchestrator exposes POST /predict — all on one public host.

## 3. Model size calculation (≤8B active)

- **Base LLM:** Qwen/Qwen3.5-4B — open-source, ≈4.0 B parameters.
- **Adapters:** 2 × LoRA (rank 32) — physics and logic — each ≈0.05 B params, *applied one at a time* (swapped per query, never summed into the active forward pass).
- **Total active parameters at any instant** = base + one adapter ≈ **4.05 B** < 8 B. Single 4B-class model active; no second LLM runs in parallel.
- Router (TF-IDF + logistic regression) and the TF-IDF retriever are non-LLM and do not count toward the limit. All components are open-source.